

B737-8 AIRCRAFT SYSTEMS

MANUAL



Caribbean
Airlines

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B737-8 AIRCRAFT SYSTEMS

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Document Control and Use

Manual Purpose

This manual explains how the major Boeing 737-8 systems are arranged, powered, controlled and monitored. It is designed to help a flight-simulation pilot understand why a switch is selected, what normal indications should appear, how systems interact, and what operational capability is lost when a component fails.

Caribbean Airlines Context

Caribbean Airlines publicly lists Boeing 737-8 and ATR 72-600 aircraft in its fleet. The Boeing 737-8 treatment in this publication is therefore a fictional Caribbean Airlines Virtual fleet concept, not a description of an announced or approved real-world subfleet.

Source and Data Hierarchy

1	The current simulator add-on FCOM, QRH/ECL, limitations and performance tool.
2	Simulator developer service bulletins and change logs.
3	Current navigation data, charts, weather, NOTAMs and virtual-operator SOP.
4	This manual for explanation, study and systems familiarization.

Information Conventions

- NORMAL describes the intended system state, not a checklist command.
- AUTO describes automatic logic that may change with phase, demand or failure.
- STANDBY describes an alternate source or path with reduced capability.
- Representative values must be checked against the modeled airframe before operational use.

CHECKLIST DISCIPLINE: Use the current simulator QRH/ECL for memory items and exact switch actions. Failure tables in this manual explain effects and priorities; they do not replace an approved checklist.

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CARIBBEAN AIRLINES VIRTUAL | B737-8 COMPLETE SYSTEMS MANUAL

1. Aircraft General and System Integration

Purpose and Architecture

The 737-8 is a twin-engine, narrow-body transport retaining the established 737 distributed system architecture while adding MAX-family displays, LEAP-1B engines, updated flight-control logic and revised electronic monitoring. Most essential functions have two independent sources, an alternate source, or a manual backup.

Basic flow: power source > distribution bus/manifold > controlled component > sensor/indication

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CREW FOCUS: Always identify both the normal source and the remaining backup before removing a generator, pump, bleed source or computer from service.

System Dependency Map

Electrical	Engine generators, APU generator, battery, TR units	Pumps, avionics, ignition, lights, controls and indications
Hydraulic	Engine-driven and electric pumps	Flight controls, gear, steering, brakes, reversers
Pneumatic	Engine bleeds, APU bleed, external air	Starts, packs, pressurization and anti-ice
Air data / IRS	Pitot-static, AOA, temperature, inertial sensors	Displays, autoflight, FMS, warnings and flight-control logic
Fuel	Tank pumps, valves and indication	Engines, APU, balance and performance
Avionics	Electrical power, cooling and sensor data	Navigation, communication, automation and alerting

Redundancy Philosophy

Separate left and right sources reduce the probability of one failure removing a complete function.

Bus ties, crossfeeds and isolation valves allow controlled sharing but can also propagate a fault if used incorrectly.

Standby systems protect only selected essential functions and may impose operational limitations.

Alerts identify a condition; they do not always identify the failed component. Crew diagnosis follows the checklist, not guesswork.

2. Flight Deck Displays and Controls

Purpose and Architecture

Four large display units present primary flight, navigation, engine and system information. Display electronic units and control panels permit reversion if a normal display path fails. The mode control panel, electronic flight instrument controls, CDU and overhead panels form the primary crew interfaces.

Basic flow: sensors/computers > display electronics > display unit > crew cross-check

CREW FOCUS: Read the Flight Mode Annunciations before interpreting the flight-director command. Selected values alone do not show what the automatic flight system is actually doing.

Display Functions

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Primary Flight Display	Attitude, airspeed, altitude, vertical speed, heading/track, flight director, FMA and alert flags.
Navigation Display	Route, map, weather radar, traffic, terrain, navigation aids and vertical profile as selected.
Engine / System	N1, N2, EGT, fuel flow, oil data, vibration, fuel quantity and system alerts.
Standby Display	Independent attitude, airspeed and altitude reference for reversion.

Controls and Indications

Display select/source controls	Select normal or alternate display sources and formats.
EFIS control panel	Sets map mode/range, barometric reference, minima and overlays.
Brightness controls	Adjust display, panel and flood lighting; very low settings can mimic display failure.
Master caution / recall	Draws attention to system annunciators and recalls stored caution states.

Failure Recognition and Operating Effects

DISPLAY SOURCE / CONTROL PANEL alert	One source or control path may be lost; use reversion and compare remaining data.
Blank display unit	Could be power, brightness, display electronics or the unit itself; preserve a valid primary-flight presentation.
Comparators / flags	Disagreement or invalid source; cross-check standby and raw sources before accepting automation.

3. Air Data, Inertial Reference and Sensors

Purpose and Architecture

Air data computers use pitot, static and temperature inputs; inertial reference units provide attitude, heading, acceleration and position. Angle-of-attack vanes, radio altimeters and other sensors feed displays, autoflight, warnings and control laws.

Basic flow: external probes > air data / IRS processing > users and comparators > flight-deck indications

CREW FOCUS: Protect probes from contamination and heat them as prescribed. A single bad input can create several secondary alerts, so identify common-source symptoms.

Air Data Sources

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Pitot probes	Dynamic pressure and computed airspeed/Mach.
Static ports	Static pressure for altitude, vertical speed and airspeed computation.
Total air temperature probe	Temperature used for air data, performance and engine calculations.
AOA vanes	Angle-of-attack input to stall warning and flight-control functions.
Radio altimeters	Height above terrain for callouts, GPWS, autoland logic and system scheduling.
IRS	Attitude, heading, track, acceleration, groundspeed and inertial position.

IRS Alignment and Position

Park the aircraft and establish stable electrical power.

Select the modeled alignment mode and enter a verified present position.

Do not move the aircraft during alignment.

After NAV is available, compare headings and position against known references.

Monitor drift and position updating during flight.

Failure Recognition and Operating Effects

IAS / ALT / AOA DISAGREE	Related sources differ beyond comparator tolerance; automation and warnings may be unreliable.
IRS fault or ALIGN light abnormal	Attitude/heading or navigation capability may be degraded; use the valid unit and checklist.
Unreliable airspeed pattern	Conflicting indications, unusual flight-director commands or warnings; establish known pitch/thrust and apply the QRH.

4. Electrical Power

Purpose and Architecture

The electrical system distributes 115-volt AC, 28-volt AC and 28-volt DC power. Each engine-driven integrated drive generator normally powers its side. The APU generator or external power can energize the system on the ground. Transformer rectifier units convert AC to DC; the battery and static inverter protect essential loads.

Basic flow: generator or external source > AC transfer buses > TR units / DC buses > essential and equipment loads

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CREW FOCUS: Before disconnecting a source, verify the replacement source is connected and stable. A lit source-available light does not always mean the source is carrying load.

Major Sources and Buses

Engine generators 1 and 2	Normal in-flight AC sources; each normally supplies its transfer bus.
APU generator	Ground or in-flight alternate source within operating envelope.
External power	Ground AC source accepted only when quality and connection logic permit.
Transfer buses	Main AC distribution paths with automatic transfer/tie logic.
TR units	Convert AC power to DC for DC buses and battery charging.
Battery / hot battery bus	Provides stored energy and continuously powered essential circuits.
Standby power	Supplies selected essential AC/DC loads after loss of normal sources.

Controls and Indications

BAT switch	Connects battery to the battery bus and permits normal charging/protection logic.
GEN switches	Reset/connect generator control breakers; spring-loaded positions discourage leaving a command selected.
BUS TRANS switch	Normally AUTO, permitting transfer and bus-tie logic.
STANDBY POWER	AUTO normally provides automatic standby transfer; OFF removes standby protection.
Ground power switch	Connects an acceptable external source to the airplane buses.

Normal Power Transitions

External power to APU: start APU, verify generator available, then connect the APU generator before removing external power.

APU to engines: after engine generators are available, connect them and confirm normal indications before stopping the APU.

A source transfer may cause momentary display or lighting changes; continuing abnormal indications require checklist action.

Failure Recognition and Operating Effects

GEN OFF BUS	Generator is not connected to its transfer bus; the opposite source or APU may carry both buses.

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SOURCE OFF	Selected transfer bus is not powered by its normal source.
TRANSFER BUS OFF	Bus is unpowered; multiple downstream equipment losses are expected.
ELEC light / battery discharge	DC charging or power-quality fault; battery endurance may become time-limiting.

5. Hydraulic Power

Purpose and Architecture

Hydraulic systems A and B each use an engine-driven pump and an electric motor-driven pump. A standby system supports selected critical functions. Reservoirs are pressurized to ensure pump inlet pressure and reduce cavitation.

Basic flow: reservoir > engine/electric pump > pressure module > actuator or brake user

CREW FOCUS: A normal pressure indication does not guarantee adequate fluid quantity. Monitor both pressure and quantity, and avoid repeatedly operating a leaking system.

Users by System

Primary flight controls	Shared actuators	Shared actuators
Landing gear normal operation	A	Manual extension alternate
Normal brakes	B	Alternate brakes from A
Nosewheel steering	A	Alternate steering may use B if installed/modeled
Leading-edge devices	B	Standby hydraulic extension
Thrust reversers	Dedicated users according to side/configuration	Opposite system/standby assistance depends on design

Controls and Indications

ENG pump switches	Control depressurization/solenoid logic; engine-driven pump turns mechanically with engine.
ELEC pump switches	Power electric motor-driven pumps for ground and backup use.
LOW PRESSURE lights	Pump output pressure below threshold.
System pressure/quantity indications	Show manifold pressure and reservoir quantity; trends matter more than a single glance.

Power Transfer Unit

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The power transfer unit can use hydraulic pressure from one system to drive a pump that pressurizes a selected function in the other system without transferring fluid between reservoirs. Its automatic logic supports high-lift operation in specified low-pressure conditions.

Failure Recognition and Operating Effects

LOW PRESSURE at one pump	Other pump may maintain the system; assess cause and demand.
Both pumps low in one system	Loss of multiple users and possible fluid leak; configure using QRH.
LOW QUANTITY / falling quantity	Leak likely; continued pump operation can worsen fluid loss.
Standby hydraulic operation	Reduced, function-specific backup—not a complete third hydraulic system.

6. Fuel

Purpose and Architecture

Fuel is carried in left and right main tanks and a center tank. AC-powered pumps supply pressure to engine and APU fuel manifolds. A crossfeed valve connects left and right manifolds. Scavenge and transfer features depend on airframe configuration and are largely automatic.

Basic flow: tank > boost pump > feed manifold/crossfeed > engine or APU

CREW FOCUS: Use crossfeed to execute an approved balance or failure procedure, not to hide an unexplained imbalance. Confirm which pumps and manifolds are actually pressurized.

Tank and Feed Logic

Main tanks 1 and 2	Primary engine feed tanks; quantity should remain balanced within limits.
Center tank	Additional fuel normally used first when pump pressure is available.
Boost pumps	Provide positive feed pressure and vapor suppression.
Crossfeed valve	Connects engine feed manifolds for single-source feeding or balance procedures.
Spar / engine fuel valves	Stop fuel at the wing/engine for shutdown or fire isolation.
Fuel temperature indication	Supports cold-fuel monitoring and operational limits.

Normal Consumption Sequence

Center-tank fuel is normally consumed before main-tank fuel when center pumps are operating and producing pressure.

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Center pumps are managed according to actual quantity and low-pressure logic; do not leave them on against an empty tank.

Main pumps remain configured for reliable engine feed.

Periodic quantity checks compare actual burn, planned burn and left/right balance.

Controls and Indications

Fuel pump switches	Command tank boost pumps; LOW PRESSURE shows inadequate pump outlet pressure.
CROSSFEED selector	Opens/closes the manifold connection; valve position light shows transit/disagreement.
Fuel quantity panel	Displays tank and total quantity; verify units match dispatch/load manager.
Engine start levers / fire switches	Control engine fuel shutoff through separate valve logic.

Failure Recognition and Operating Effects

LOW PRESSURE	Pump not producing pressure, tank empty, switch/source issue or pump failure.
Fuel imbalance	Unequal consumption, pump configuration, crossfeed state or possible leak.
Fuel leak suspicion	Unexpected total loss, imbalance and fuel-flow mismatch; land before reserve is threatened.
FILTER BYPASS	Restriction/bypass condition requiring maintenance and operational assessment.

7. Pneumatic and Bleed Air

Purpose and Architecture

High-temperature compressed air comes from engine compressor stages, the APU or external ground air. The bleed manifold supplies engine start, air-conditioning packs, pressurization, wing/engine anti-ice and selected services. Isolation and pressure-regulating valves control source sharing.

Basic flow: engine/APU/external air > bleed regulating valve > manifold/isolation valve > packs, starts and anti-ice

CREW FOCUS: Pneumatic configuration is a demand-management problem. Engine start, anti-ice and two packs can compete for pressure, so follow the modeled configuration exactly.

Bleed Sources and Users

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Engine bleed	Variable-stage compressor air regulated for manifold use.
APU bleed	Primary ground source for engine start and packs; available in flight within limits.
External air	Ground cart supplies start or conditioning air through designated connection.
Packs	Largest continuous pneumatic consumers in normal operation.
Engine start valves	Route manifold air to the air-turbine starters.
Thermal anti-ice	Uses hot bleed air at engine cowls and wing leading edges.

Controls and Indications

BLEED switches	Command associated bleed-air valve logic.
ISOLATION VALVE	AUTO uses pack/bleed configuration logic; OPEN/CLOSE overrides.
PACK switches	OFF/AUTO/HIGH schedules pack flow and demand.
DUCT PRESS indication	Displays left/right manifold pressure for configuration and fault recognition.

Failure Recognition and Operating Effects

BLEED TRIP OFF	Overtemperature/overpressure protection closes the engine bleed.
WING-BODY OVERHEAT	Leak/overheat detection in pneumatic zones; isolate sources as checklist directs.
Low duct pressure	Source unavailable, valve closed, leak or excessive demand.
Start valve open indication	Valve position inconsistent with expected start cycle; avoid starter damage.

8. Air Conditioning and Equipment Cooling

Purpose and Architecture

Two air-conditioning packs cool and condition pneumatic air. Temperature-control systems mix pack air and warmer trim air for flight deck, forward cabin and aft cabin zones. Recirculation fans reduce bleed demand. Equipment cooling fans move air through avionics racks.

Basic flow: bleed manifold > pack cooling cycle > mix manifold/trim air > flight deck and cabin zones

CREW FOCUS: A comfortable cabin does not prove pressurization is normal. Monitor pack state, zone temperatures and cabin pressure as separate systems.

Pack Cooling Process

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Bleed air enters the pack through a flow-control valve.

Heat exchangers reject heat to ram air.

Air-cycle machinery compresses, cools and expands the air.

Water separation reduces moisture.

Conditioned air enters the mix manifold and is distributed by zone.

Controls and Indications

PACK switches	Select off, automatic flow or high flow depending on configuration.
Temperature selectors	Command desired zone temperatures through automatic controllers.
TRIM AIR switch	Enables trim-air pressure regulation and zone temperature valves.
RECIRC FAN switches	Reduce pack demand by returning filtered cabin air to the mix manifold.
Equipment cooling selectors	Control supply/exhaust fan operation for avionics cooling.

Failure Recognition and Operating Effects

PACK light / trip	Pack overheat or control fault; one pack can pressurize with capacity limitations.
ZONE TEMP light	Controller/valve fault or duct overheat; automatic temperature control may be lost.
EQUIP COOLING OFF light	Insufficient avionics cooling airflow; equipment reliability may become time-limiting.
Smoke/fumes from conditioning	Possible pack, oil or electrical source; use smoke/fumes checklist and consider immediate landing.

9. Cabin Pressurization

Purpose and Architecture

The airplane is pressurized by conditioned pack air. Cabin pressure is controlled primarily by modulating the outflow valve; safety/relief valves protect against excessive positive or negative differential pressure. Two automatic controllers provide normal and alternate control, with manual control available.

Basic flow: pack inflow > pressurized fuselage > outflow valve controller > scheduled cabin altitude

CREW FOCUS: Set and verify landing altitude before departure. The controller cannot schedule a correct landing if the destination elevation is wrong.

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Automatic Schedule

On the ground, the airplane is slightly depressurized for door operation.

During takeoff, the system may prepressurize to reduce pressure transients.

In climb, cabin altitude rises at a controlled rate to a scheduled cruise cabin altitude.

In descent, the cabin descends toward the selected landing altitude and slightly pressurizes before touchdown.

Controls and Indications

Flight altitude selector	Provides planned cruise altitude to the controller on applicable panel versions.
Landing altitude selector	Sets destination field elevation for the pressure schedule.
Mode selector	AUTO/ALTN selects automatic controller logic; MAN permits direct outflow-valve positioning.
Cabin altitude/rate/differential	Shows actual pressurization performance, not merely controller command.

Failure Recognition and Operating Effects

AUTO FAIL	Selected automatic controller has failed or detected a fault; alternate mode may control.
ALTN light	Alternate automatic controller is active.
CABIN ALTITUDE warning	Cabin altitude exceeds warning threshold; oxygen and emergency descent actions may be required.
Rapid rate / differential abnormal	Leak, valve or controller issue; confirm trend and prepare for descent/diversion.

10. Ice and Rain Protection

Purpose and Architecture

Electrical heat protects pitot-static probes, AOA sensors, total-air-temperature probe and flight-deck windows. Hot bleed air protects engine cowl lips and selected wing leading-edge slats. Windshield wipers and rain-repellent features depend on configuration.

Basic flow: electrical or bleed source > control/protection logic > heated surface > ice prevention or shedding

CREW FOCUS: Use anti-ice based on the published definition of icing conditions, not only after ice becomes visible. Anti-ice changes bleed demand and performance.

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Protected Areas

Pitot / static / AOA / TAT	Electrical probe heat; automatic/normal logic varies with aircraft configuration.
Flight-deck windows	Electrical conductive heat for anti-ice and structural strength.
Engine inlet cowls	Engine bleed-air thermal anti-ice.
Wing leading edges	Bleed-air thermal anti-ice to selected slats, normally in flight.
Windshields	Wipers and heated windows; rain repellent if installed.

Controls and Indications

PROBE HEAT switches	Command or arm probe-heating circuits according to panel logic.
WINDOW HEAT switches	Power four window heat zones; ON lights confirm command, not surface condition.
ENGINE ANTI-ICE switches	Open cowl anti-ice valves and modify engine/bleed scheduling.
WING ANTI-ICE switch	Commands wing thermal anti-ice subject to ground/in-flight logic.
Wiper selectors	Select speed; park before the windshield dries to avoid damage in realistic simulation.

Failure Recognition and Operating Effects

ANTI-ICE valve light bright	Valve position disagrees or is in transit.
COWL / WING anti-ice alert	Pressure/temperature or valve fault may leave the protected area unheated.
WINDOW OVERHEAT	Controller detects excessive temperature; affected window heat is removed.
Probe heat failure	Related air data may become unreliable in moisture/icing.

11. Flight Controls, Speed Trim and MCAS

Purpose and Architecture

Primary pitch, roll and yaw control is conventional and hydraulically powered through dual control paths. Secondary controls include the stabilizer, leading-edge devices, trailing-edge flaps, spoilers/speedbrakes and yaw damper. Electronic augmentation improves handling while preserving pilot control and mechanical/manual backups.

Basic flow: pilot controls / computers > cables and hydraulic actuators > control surfaces > aircraft response and feedback

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CREW FOCUS: Unexpected trim or control motion demands immediate flight-path control and exact checklist action. Do not troubleshoot by randomly cycling cutout switches.

Primary Controls

Pitch	Elevators powered by hydraulic systems A and B	Manual reversion; stabilizer trim; speed trim and MCAS functions.
Roll	Ailerons and flight spoilers	Manual aileron reversion with increased forces; spoiler mixing.
Yaw	Rudder power control unit	Standby rudder system; yaw damper and stall-related augmentation.

Stabilizer Trim

Electric trim is commanded by control-wheel switches and automatic functions. The autopilot also commands stabilizer trim. Manual trim wheels provide a mechanical method, although aerodynamic loads can make manual movement difficult. Main electric and autopilot trim cutout architecture permits electrical isolation as defined by the current procedure.

Speed Trim System

The speed trim system can command stabilizer movement during specified low-speed, high-thrust manual-flight conditions to improve speed stability. It operates through flight-control computers and is distinct from autopilot trim, though both use the electric stabilizer system.

Maneuvering Characteristics Augmentation System

MCAS is a flight-control law within the speed trim system intended to improve pitch handling in a specific manual-flight, flaps-up, elevated-angle-of-attack region. Boeing states that updated logic compares both AOA sensors; if they disagree by 5.5 degrees or more with flaps retracted, MCAS does not activate and an AOA DISAGREE indication alerts the crew. For a qualifying elevated-AOA event, updated MCAS logic provides a single input and limits authority so the crew can counteract the command.

MCAS AWARENESS: MCAS has no dedicated cockpit switch. Pilots respond to aircraft motion and trim indications using the applicable stabilizer/non-normal procedure, not by attempting to diagnose MCAS in real time.

High-Lift and Spoiler Systems

Trailing-edge flaps are hydraulically driven through a transmission; alternate electrical extension provides reduced capability.

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Leading-edge flaps/slats extend with trailing-edge flap command and have standby hydraulic extension capability.

Flight spoilers supplement roll control; ground spoilers deploy for landing/rejected takeoff when arming and logic conditions are satisfied.

Speedbrake use increases drag and can affect buffet margins; retract before landing unless procedure directs otherwise.

Failure Recognition and Operating Effects

FEEL DIFF PRESS	Elevator feel-system pressure mismatch; control feel may be abnormal.
MACH TRIM / SPEED TRIM fail	Automatic handling augmentation unavailable; follow checklist and operating restrictions.
AOA DISAGREE	Left/right AOA mismatch; MCAS inhibited under updated logic and related displays/warnings may be affected.
Flap asymmetry/disagree	High-lift motion stops or indication disagrees; do not reselect casually.
Stabilizer trim runaway/uncommanded	Control pitch, hold column, use exact memory items/checklist and plan landing.

12. Automatic Flight

Purpose and Architecture

Two flight-control computers support flight directors, autopilot, autothrottle and mode logic. The mode control panel provides selected targets; FMS guidance provides managed targets. Automatic flight commands are displayed on the FMA.

Basic flow: MCP/FMS targets > flight-control computers > autopilot/flight director/autothrottle > control surfaces and thrust

CREW FOCUS: Every automation selection is a three-step action: command, verify FMA, monitor aircraft response.

Major Modes

Lateral	HDG SEL, LNAV, VOR/LOC, APP/ROLLOUT as equipped.
Vertical	LVL CHG, V/S, ALT HOLD, VNAV, G/S, FLARE as equipped.
Thrust	N1, SPEED, ARM, RETARD/idle logic depending on phase.
Approach	Single- or dual-channel logic; autoland capability depends on aircraft status and airport/crew approval.

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Controls and Indications

MCP speed/heading/altitude selectors	Set immediate targets and enable selected-mode control.
A/P engage/disengage	Engages a command channel when conditions permit; disengagement produces warnings.
A/T ARM switch	Arms autothrottle servo control but does not guarantee an active thrust mode.
TO/GA switches	Command takeoff or go-around guidance according to phase and configuration.

Failure Recognition and Operating Effects

Unexpected FMA change	Automation has changed strategy; intervene with a simpler mode or disconnect.
A/P disengage warning	Autopilot no longer controls; take manual control and cancel warning deliberately.
A/T disengage / thrust asymmetry	Manually set and monitor thrust; verify mode and servo status.
Dual-channel approach downgrade	Autoland capability reduced; continue only if minima and briefing remain valid.

13. Flight Management System

Purpose and Architecture

The FMS integrates navigation data, route definition, performance predictions, guidance targets and progress monitoring. Two CDUs provide crew access; flight-management computers use IRS, GNSS and radio updating to calculate position and guidance.

Basic flow: navigation database and crew entries > FMC position/performance calculation > LNAV/VNAV guidance > displays and automatic flight

CREW FOCUS: A route displayed in magenta is not automatically correct. Verify every leg, constraint and discontinuity against clearance and charts.

Core Functions

Route management	Origin, destination, airways, fixes, procedures, runways and discontinuities.
Performance	Weights, reserves, cost index, cruise level, thrust limits and speed schedule.
Navigation	Blended aircraft position and required/actual navigation performance.
Guidance	LNAV lateral path, VNAV climb/cruise/descent path and target speeds.
Prediction	Fuel, ETA, altitude and speed at route points.

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Programming Discipline

- Verify IDENT page aircraft model and navigation-database cycle.
- Enter and verify origin, destination, route and flight number.
- Select procedures from the clearance and charts.
- Enter performance data with correct units.
- Calculate/enter thrust and takeoff data using the performance source.
- Step through the entire route on the map.
- Cross-check PROGRESS fuel/ETA against the operational flight plan during flight.

Failure Recognition and Operating Effects

FMC message	Condition needs crew attention; read scratchpad and associated page before clearing.
UNABLE NEXT ALT / path message	Planned profile cannot satisfy a constraint under current conditions.
Position uncertainty / RNP exceeded	Navigation performance insufficient; revert to approved sensors/procedure or discontinue approach.
Database expired or procedure absent	Do not invent procedure coding; use current data or an approved raw-data method.

14. Navigation

Purpose and Architecture

Navigation combines inertial, GNSS and radio sources with weather radar, terrain awareness, traffic systems, radio altimeters and magnetic heading references. Source selection and comparator monitoring protect against accepting a single invalid solution.

Basic flow: external/inertial sensor > navigation receiver/computer > FMS or display processing > crew and automatic-flight use

CREW FOCUS: Use independent raw data to monitor FMS guidance during departure, approach and any operation near terrain.

Navigation Equipment

VOR/ILS receivers	Conventional navigation, localizer and glideslope guidance.
DME	Slant-range distance and FMS radio updating.

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GNSS	High-accuracy position input subject to integrity/availability.
Weather radar	Precipitation reflectivity and turbulence-related information; not terrain mapping.
EGPWS / terrain display	Terrain/obstacle awareness and escape warnings.
TCAS	Traffic display and resolution/advisory guidance.
Radio altimeters	Low-altitude height input for warnings, callouts and autoland.

Weather Radar Technique

Use tilt and range actively to understand cell structure, not simply leave one setting all flight.

Strong attenuation can hide weather beyond a heavy return.

Avoid using gaps between intense returns as guaranteed safe corridors.

Ground clutter can confirm radar operation but is not a substitute for a test procedure.

Failure Recognition and Operating Effects

NAV source flag	Selected radio or data source invalid.
TERR / GPWS alert	Respond immediately according to warning type and procedure; do not delay for diagnosis.
TCAS RA	Follow RA vertical guidance unless doing so would endanger the aircraft.
Radar failure	Use external weather sources and conservative avoidance; do not penetrate suspected cells.

15. Communications and Surveillance

Purpose and Architecture

VHF radios provide primary voice communication. Audio control panels route transmit and receive sources. The transponder supplies surveillance data and supports TCAS. SELCAL, datalink, SATCOM or HF capability depends on the installed simulator configuration.

Basic flow: crew audio/control panel > radio or datalink unit > antenna/network > ATC, company or aircraft receiver

CREW FOCUS: File only equipment and surveillance codes actually supported by the simulator aircraft and network client.

Audio and Radio Management

Audio control panel	Select transmit source, receive sources, volume, interphone and speaker.
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VHF control panel	Tune active/standby frequencies and transfer selections.
Transponder panel	Select code and operating mode; altitude reporting/TCAS state depends on mode.
Cockpit voice recorder	Records flight-deck audio subject to power and operating logic.
Emergency locator transmitter	Provides distress transmission when activated automatically or manually, if modeled.

TCAS Logic

Traffic advisories aid visual acquisition and prepare the crew.

Resolution advisories command vertical guidance and require prompt response.

Do not maneuver solely from traffic display bearing without an RA or visual/ATC basis.

Select TA-only when required by abnormal procedure or operational condition.

Failure Recognition and Operating Effects

Radio failure	Try alternate radio/audio path, verify volume/squelch, use network text or transponder procedures.
Audio panel failure	Use alternate panel/speaker/hand microphone path as modeled.
Transponder / TCAS fail	Surveillance or collision-avoidance capability reduced; advise ATC and observe restrictions.

16. Landing Gear and Brakes

Purpose and Architecture

Hydraulic system A normally retracts and extends the landing gear. Mechanical uplocks hold the main gear; doors and sequencing are coordinated by gear logic. Manual extension releases uplocks for gravity/aerodynamic extension. Hydraulic braking includes antiskid, autobrake, alternate braking and parking brake functions.

Basic flow: gear lever and hydraulic pressure > selector valve/actuator/uplocks > gear position sensors > lights and air/ground logic

CREW FOCUS: Three green lights confirm down-and-locked indications, but a safe landing decision must also consider door, brake, steering and tire status.

Gear Operation

After takeoff, gear retraction removes drag and automatically stops hydraulic flow when the sequence is complete.

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For landing, normal extension uses hydraulic pressure; manual extension provides an alternate release path.

Air/ground sensors control many unrelated systems, so a sensing fault can create multiple unusual indications.

Brake System

Normal brakes	System B pressure through metering and antiskid valves.
Alternate brakes	System A pressure when normal braking unavailable; antiskid capability depends on failure.
Autobrake	Commands programmed deceleration for takeoff rejection or landing.
Antiskid	Detects wheel deceleration/skid and reduces pressure to preserve tire traction.
Parking brake	Traps metered brake pressure mechanically when set.
Brake temperature monitoring	Supports cooling assessment and fuse-plug/tire risk management.

Controls and Indications

Landing gear lever	Commands UP/OFF/DOWN sequence; lock override exists for specified ground anomalies.
Gear indicator lights	Green = down and locked; red indicates unsafe/disagree conditions.
AUTOBRAKE selector	RTO or landing deceleration level; DISARM light shows system not armed/available.
Antiskid switch	Normally ON; OFF changes braking technique and stopping performance.

Failure Recognition and Operating Effects

Gear disagree / red light	Commanded and sensed position differ; use checklist and alternate extension as required.
ANTISKID INOP	Wheel-lock protection reduced; landing performance and braking technique change.
AUTO BRAKE DISARM	Automatic deceleration unavailable or released; use manual braking.
Hot brakes	Fire/tire risk and turnaround restriction; avoid setting parking brake if procedure directs.

17. CFM LEAP-1B Powerplant

Purpose and Architecture

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MANUAL

Two CFM International LEAP-1B high-bypass turbofan engines provide thrust, electrical generation, hydraulic pump drive and bleed air. Electronic engine controls schedule fuel, variable geometry, starting and protection functions using thrust-lever commands and sensor inputs.

Basic flow: inlet/fan/compressors > combustor > high/low pressure turbines > exhaust thrust and accessory power

CREW FOCUS: Monitor trend and symmetry, not just limits. A slowly changing oil, vibration or temperature indication can be more significant than one normal snapshot.

Engine Sections and Indications

N1	Low-pressure rotor/fan speed; primary thrust-setting reference.
N2	High-pressure rotor speed; critical during start and accessory operation.
EGT	Exhaust gas temperature; start margin and engine health indicator.
Fuel flow	Instantaneous fuel consumption; useful for symmetry and planning.
Oil pressure / temperature / quantity	Lubrication condition and usable oil state.
Vibration	Rotor vibration trend; abnormality may indicate damage or imbalance.

Starting Sequence

- Provide pneumatic start air and electrical control power.
- Move the start switch to GND; starter opens and N2 increases.
- At the modeled minimum N2, move the start lever to IDLE.
- Fuel and ignition produce light-off; monitor EGT, N1, N2, fuel flow and oil pressure.
- Starter cuts out automatically near self-sustaining speed; verify stable idle.

Electronic Engine Control

- The EEC translates thrust-lever position and environmental/aircraft inputs into engine commands.
- Rated and derated thrust modes are shown through thrust-limit indications.
- Protection logic is not permission to ignore exceedances; the crew still monitors limits and responds to abnormal indications.
- Alternate EEC modes may provide reduced optimization and require maintenance/dispatch action.

Failure Recognition and Operating Effects

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Hot start	EGT rises beyond start limit; cut fuel and motor the engine as procedure directs.
Hung start	N2/EGT stabilizes below normal idle; stop start to prevent damage.
No light-off	Fuel introduced but no normal EGT rise; cut fuel and clear fuel vapors per checklist.
Engine flameout/failure	Loss of thrust with indication changes; control aircraft, identify/confirm and use QRH.
Severe damage / separation / fire	Prompt isolation and nearest-suitable landing; do not delay for diagnosis.

18. Auxiliary Power Unit

Purpose and Architecture

The APU is a small gas-turbine powerplant in the tail providing electrical power and pneumatic air. Its electronic control unit manages start, speed, protection and shutdown. Availability depends on altitude and demand.

Basic flow: fuel and battery/start power > APU compressor/turbine > generator and bleed valve > aircraft buses/manifold

CREW FOCUS: Allow the modeled cooldown after high pneumatic load before shutdown. Immediate shutdown is reserved for fire or checklist-directed urgency.

Controls and Indications

APU switch OFF/ON/START	Initiates start sequence, run command and normal shutdown.
APU GEN switches	Connect APU generator to one or both transfer buses as system logic permits.
APU BLEED switch	Connects APU pneumatic source to the bleed manifold.
EGT and maintenance indications	Show start/run condition and protective shutdown cues.

Normal Uses

Ground electrical power before engine start and after engine shutdown.

Pneumatic air for air conditioning and engine start.

In-flight alternate electrical or pneumatic source within operating envelope.

Backup source during single-generator or bleed abnormalities.

Failure Recognition and Operating Effects

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APU GEN OFF BUS	Generator is available or disconnected depending on associated indications; verify connection/load.
APU fault / protective shutdown	Electronic control detected an unsafe condition and stopped the unit.
APU fire warning	Execute fire shutdown/extinguishing actions and evacuate/coordinate ground response as required.
No start / no EGT rise	Start sequence failed; observe restart and maintenance limitations.

19. Fire and Overheat Protection

Purpose and Architecture

Detection loops monitor engines, APU, wheel well and pneumatic zones. Cargo compartments use smoke detection and extinguishing/suppression equipment according to configuration. Engine and APU fire switches isolate fuel, bleed, generator and hydraulic-related services while arming extinguishing bottles.

Basic flow: detector loop > fire/overheat logic > warning and master warning > isolation/extinguishing action

CREW FOCUS: A warning means act first to protect the aircraft, then confirm and complete the checklist. Do not continue flight to diagnose a persistent or uncontained fire.

Protected Zones

Engines	Dual-loop fire/overheat detection and extinguishing bottles.
APU	Fire detection, automatic/crew shutdown logic and extinguisher.
Cargo compartments	Smoke detection and fixed extinguishing/suppression capability.
Wheel well	Fire/overheat detection; normally no built-in extinguishing.
Wing/body pneumatic zones	Overheat detection for bleed-air leaks.
Lavatories	Smoke detection and localized waste-bin extinguishing equipment.

Controls and Indications

Fire warning test	Checks warning lights, bells and detection logic as modeled.
Engine fire switches	When unlocked/pulled, isolate associated engine services and arm extinguisher circuits.
APU fire switch	Shuts down/isolate APU and arms extinguisher.

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Extinguisher discharge

Rotating/pushing command discharges selected bottle; indications show circuit/bottle status.

Failure Recognition and Operating Effects

Fire warning	Immediate memory action/checklist; land as soon as appropriate for persistence/severity.
Overheat without fire warning	Temperature/leak condition may precede fire; isolate source and monitor.
Cargo smoke	Begin suppression and plan landing; extinguishing agent duration affects diversion choice.
Detector fault	Protection coverage degraded; dispatch/operating restrictions may apply.

20. Warning and Alerting Systems

Purpose and Architecture

Warnings combine lights, sounds, voice alerts, stick shaker and display indications. Master warning identifies immediate-attention conditions; master caution directs attention to system annunciators. GPWS, TCAS, takeoff configuration and cabin-altitude warnings use dedicated logic.

Basic flow: sensor/logic condition > aural/visual alert > crew recognition > memory item/checklist and flight-path response

CREW FOCUS: Respond to the highest-priority warning first. Multiple cautions often share one failed power, sensor, hydraulic or pneumatic source.

Alert Priority

Warning	Fire, cabin altitude, takeoff configuration, overspeed, GPWS: immediate recognition and action.
Caution	System fault requiring prompt attention after flight path is safe.
Advisory / status	Changed system state, degradation or configuration information.
Comparator flag	Conflicting or invalid data; cross-check independent sources.

Takeoff Configuration Warning

With thrust advanced for takeoff, the warning logic monitors critical configuration such as flaps, stabilizer trim, speedbrake and parking brake. A warning requires immediate thrust reduction and correction; do not attempt to identify the item while accelerating.

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Stall Warning

Control-column stick shaker provides tactile warning based on AOA and configuration logic. It requires immediate stall-recovery response while respecting terrain and configuration. Sensor disagreement can affect warning behavior; fly reliable attitude and thrust references.

Failure Recognition and Operating Effects

Master warning	Immediate-attention event; identify dedicated warning while maintaining control.
Master caution	Review system annunciator/display and complete checklist when safe.
Nuisance or contradictory alert	Treat as real until cross-check proves otherwise; use independent instruments and QRH.
Alert flood	Look for one upstream failure and prioritize flight path, fire, pressurization and terrain.

21. Oxygen

Purpose and Architecture

A gaseous crew oxygen system supplies flight-deck masks. Passenger oxygen is supplied by chemical generators or gaseous equipment depending on configuration and deploys automatically at a specified cabin altitude or manually from the flight deck. Portable bottles support cabin response.

Basic flow: oxygen source > shutoff/regulator > mask or passenger unit > user breathing supply

CREW FOCUS: Know mask location and operation before flight. During depressurization, don the mask before troubleshooting or communicating.

Flight-Deck Oxygen

Pressure/quantity is checked during preflight against dispatch requirements.

Quick-donning masks provide normal, 100-percent and emergency/pressure modes according to design.

Mask microphones require correct audio-panel/interphone configuration.

A crew oxygen leak can reduce endurance and may require descent/diversion.

Passenger Oxygen

Automatic deployment is based on cabin-altitude logic; manual deployment is available.

Chemical generators become hot after activation and normally cannot be stopped.

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Passenger oxygen supports emergency descent, not extended cruise at high cabin altitude.

Cabin crew coordinate mask use, portable equipment and medical events.

Failure Recognition and Operating Effects

Low crew oxygen pressure	Crew protection endurance reduced; assess flight duration/altitude and maintenance status.
Passenger masks deployed	Treat as evidence of cabin-altitude event unless confirmed otherwise; descend and investigate.
Mask communication difficulty	Use interphone/radio selections and hand signals; do not remove oxygen in unsafe cabin altitude.

22. Doors, Windows and Exits

Purpose and Architecture

Plug-type passenger and service doors use cabin pressure to strengthen closure in flight. Cargo doors, access panels and emergency exits include latches and warning switches. Flight-deck windows provide visibility, heat and emergency egress functions.

Basic flow: door/exit structure > latches and seals > proximity/position sensing > flight-deck indication

CREW FOCUS: A door light is not cleared by assumption. Verify the physical door and latch state through ground personnel or the applicable procedure.

Door Categories

Forward/aft entry doors	Plug-type passenger doors with arming/disarming and evacuation-slide interfaces.
Service doors	Cabin service access with similar latch/indication principles.
Cargo doors	Externally operated doors with mechanical latches and position warning.
Overwing exits	Emergency exits with flight-lock/indication features depending on configuration.
Flight-deck windows	Heated panes; selected windows open for ground ventilation/emergency egress.
Equipment/access panels	Ground-only access; some monitored by door warning system.

Failure Recognition and Operating Effects

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DOORS light	One or more monitored doors/panels not confirmed closed; identify on annunciation panel/display.
Abnormal noise / pressurization leak	Possible seal or structural issue; stop climb, assess cabin and use checklist.
Window heat failure or damage	Structural/visibility protection reduced; observe speed/altitude restrictions and land as directed.

23. Lighting

Purpose and Architecture

Exterior lighting supports visibility, anti-collision signaling, runway illumination and inspection. Interior lighting supports instrument readability, emergency egress and cabin service. Independent controls and dimming circuits allow configuration by phase and ambient conditions.

Basic flow: electrical bus > switch/dimmer > light unit > visibility or signaling function

CREW FOCUS: Lighting is operational communication: beacon before engine movement, strobe for runway/flight, and landing lights for conspicuity subject to SOP.

Exterior Lighting

Anti-collision beacon	Warns ground personnel that engines may start or are operating.
Strobe	High-intensity anti-collision light for runway entry and flight.
Position/navigation	Shows aircraft orientation and presence.
Landing / runway turnoff	Forward illumination and conspicuity for takeoff, landing and taxi.
Taxi	Illuminates taxi path; avoid dazzling ground staff.
Wing / wheel-well	Inspection of leading edges, engines or gear areas.

Interior and Emergency Lighting

Panel, background and flood lighting must preserve night vision without hiding annunciators.

Emergency-exit lighting is armed for automatic illumination after power loss or impact logic, with manual ON/OFF control.

Cabin signs integrate passenger information with crew SOP.

A dimmer at minimum can be mistaken for display or panel-light failure.

24. Cabin and Passenger Systems

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MANUAL

Purpose and Architecture

Cabin systems include passenger signs, calls, interphone, public address, lavatory detection, galley electrical loads, potable/waste water equipment and optional in-flight entertainment/connectivity. These systems affect safety, workload and electrical demand even though most are cabin-managed.

Basic flow: aircraft power/air/water > cabin control or service unit > passenger-service equipment > crew/passenger use

CREW FOCUS: Coordinate before removing galley power, changing signs or shutting down packs; cabin consequences can affect evacuation readiness and passenger health.

Major Cabin Interfaces

Passenger signs	Seat-belt and no-smoking/electronic signage according to SOP and phase.
PA / interphone	Safety announcements and communication with cabin/ground crew.
Cabin call system	Normal and emergency call indications.
Galley power	Large electrical load subject to load shedding and crew isolation.
Lavatory smoke	Dedicated warning requiring immediate cabin coordination.
Cabin altitude/temperature	Passenger comfort and safety; corroborates pressurization/pack performance.

25. System Integration by Phase of Flight

Cold and Dark to Electrical Power

Confirm aircraft safety state, battery voltage and switch configuration.

Connect an acceptable external source or start the APU.

Verify transfer buses, TR/DC buses and battery charging.

Power IRS/avionics and establish equipment cooling.

Configure emergency lighting and required cabin services.

Engine Start Integration

Start energy chain: electrical control power > pneumatic pressure > starter rotation > fuel and ignition > self-sustaining idle

Fuel pumps provide positive feed.

Hydraulic electric pumps are configured to protect ground personnel and provide required pressure.

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Packs/bleeds/isolation valve are arranged for starter pressure.

Generator, bleed, hydraulic and anti-ice configuration is restored after start.

Takeoff Integration

Flight controls	Controls checked; flaps and stabilizer trim agree with performance data.
Autoflight	Flight directors on; correct modes/targets and takeoff guidance available.
Engines	Rated/derated thrust set, stable and within limits.
Warnings	No takeoff configuration warning; recall complete.
Brakes/gear	Autobrake RTO, antiskid available, gear indications normal.
Pneumatic/anti-ice	Configuration matches performance calculation.

Cruise Monitoring

Electrical source and load symmetry.

Hydraulic pressure/quantity and pump status.

Fuel burn, balance and temperature.

Engine oil, vibration, EGT and fuel-flow trends.

Cabin altitude, rate and differential pressure.

Weather, navigation performance and alternate suitability.

Approach and Landing Integration

Pressurization landing altitude and cabin descent are correct.

Landing gear, high-lift devices, speedbrake and autobrake are configured.

Radio/navigation sources and minima match the approach.

Hydraulic and brake capability support planned stopping distance.

Engine anti-ice/packs and landing performance assumptions agree.

Appendix A — Failure Effects Quick Reference

One AC transfer bus	Displays/avionics, pumps, heat and equipment on that bus may be lost; transfer logic may restore some loads.
One hydraulic system	Multiple flight-control actuators and specific gear/brake/steering users degraded; remaining system carries higher importance.

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One bleed source	Pack/anti-ice/start capability reduced; isolation and demand management become important.
One air-data source	Comparator alerts, automation degradation, possible warning/control-law effects.
One IRS	Attitude/heading/navigation redundancy reduced; select valid sources.
One engine	Thrust, generator, hydraulic pump and bleed source lost together; cross-system workload increases.
Equipment cooling	Avionics reliability time-limited; land if checklist requires.
Cabin pressurization control	Manual control or emergency descent may be required.

Appendix B — Cockpit Panel Scan

Forward overhead	Electrical, IRS, oxygen, lights, window/probe heat.
Center overhead	Fuel, hydraulics, anti-ice, pneumatics, packs, pressurization.
Aft overhead	Audio, service/interphone, engine/APU fire protection.
Glareshield/MCP	EFIS sources, speed/heading/altitude, flight directors, autopilot/autothrottle.
Main panel	PFD/ND validity, standby instruments, engine/system display, gear and brakes.
Pedestal	CDUs, radios, transponder, trim, speedbrake, thrust, flaps and parking brake.

Appendix C — Glossary and References

Glossary

AC	Alternating current
ADC	Air data computer
AOA	Angle of attack
APU	Auxiliary power unit
CDU	Control display unit
DC	Direct current
EEC	Electronic engine control
ECL	Electronic checklist
EGPWS	Enhanced ground proximity warning system
FMA	Flight mode annunciation
FMC/FMS	Flight management computer/system

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IRS	Inertial reference system
MCAS	Maneuvering Characteristics Augmentation System
MCP	Mode control panel
QRH	Quick Reference Handbook
TCAS	Traffic alert and collision avoidance system
TR unit	Transformer rectifier unit
VNAV/LNAV	Vertical/lateral navigation

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FINAL REMINDER: Use this manual to understand systems. Use the current aircraft-specific checklist and limitations to operate the simulator. Never apply this document to a real aircraft.